

Hormonal Influence on Oral Tissues

PREGNANCY GINGIVITIS

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Learning Objectives

At the end of this presentation, students will be able to:

1. Understand the role of hormones in maintaining oral health.
2. Explain the effects of hormonal changes during pregnancy on oral tissues.
3. Understand the relationship between plaque accumulation and hormonal changes in the progression of gingival conditions.
4. Differentiate pregnancy gingivitis from other forms of periodontal diseases.
5. Understand the importance of oral hygiene and dental care during pregnancy.

Introduction

- Pregnancy is associated with profound physiological and hormonal changes.
- These changes influence many body tissues including oral tissues.
- Oral manifestations are common during pregnancy because gingival tissues become more sensitive to local irritants such as plaque.
- Pregnancy gingivitis is one of the most frequently observed oral conditions in pregnant women.
- Understanding the biochemical basis of this condition is important.

Role of Hormones in Maintaining Oral Health

Hormones and Oral Tissues

Hormones regulate:

- Tissue growth and repair
- Blood circulation
- Immune response
- Bone metabolism
- Salivary gland activity

Role of Hormones in Maintaining Oral Health (Continued..)

Major Hormones Affecting Oral Health

- Estrogen
- Progesterone
- Parathyroid hormone
- Cortisol
- Insulin

Balanced hormone levels are essential for maintaining healthy gingiva and periodontal tissues.

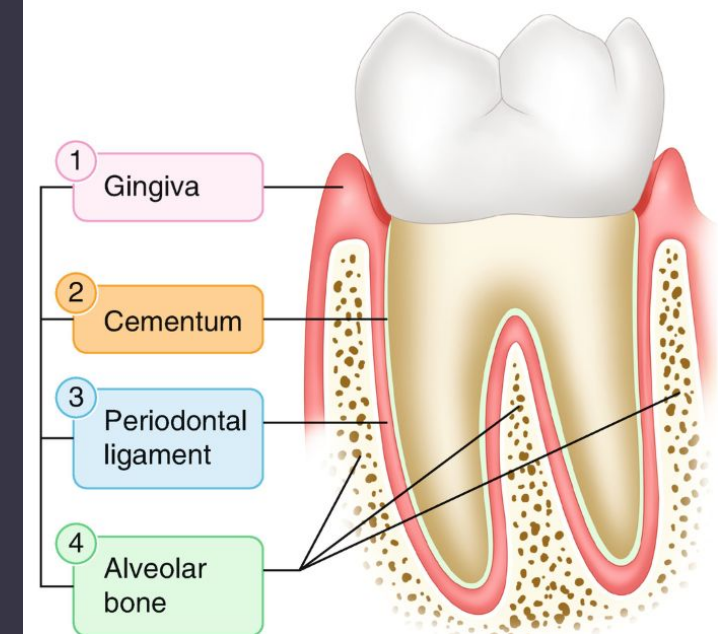
Normal Gingiva and Periodontium

Components of Periodontium

- Gingiva
- Periodontal ligament
- Cementum
- Alveolar bone

Features of Healthy Gingiva

- Pale pink color
 - Firm consistency
 - Knife-edged margins
 - No bleeding on brushing
- Healthy oral tissues depend on:
- Good oral hygiene
 - Balanced hormones
 - Proper immune function



Hormonal Changes During Pregnancy

Pregnancy Hormones

During pregnancy:

- Estrogen levels increase significantly.
- Progesterone levels may increase up to 10 times normal levels.

These hormones are produced mainly by:

- Placenta
- Ovaries

Hormonal levels rise progressively and peak during the third trimester.

Effects of Estrogen on Oral Tissues

Estrogen

Actions on Gingiva

- Increases vascular proliferation
- Enhances capillary permeability
- Stimulates inflammatory response
- Affects epithelial maturation

Clinical Effects

- Gingival redness
- Increased bleeding tendency
- Increased tissue sensitivity

Effects of Progesterone on Oral Tissues

Progesterone

Effects

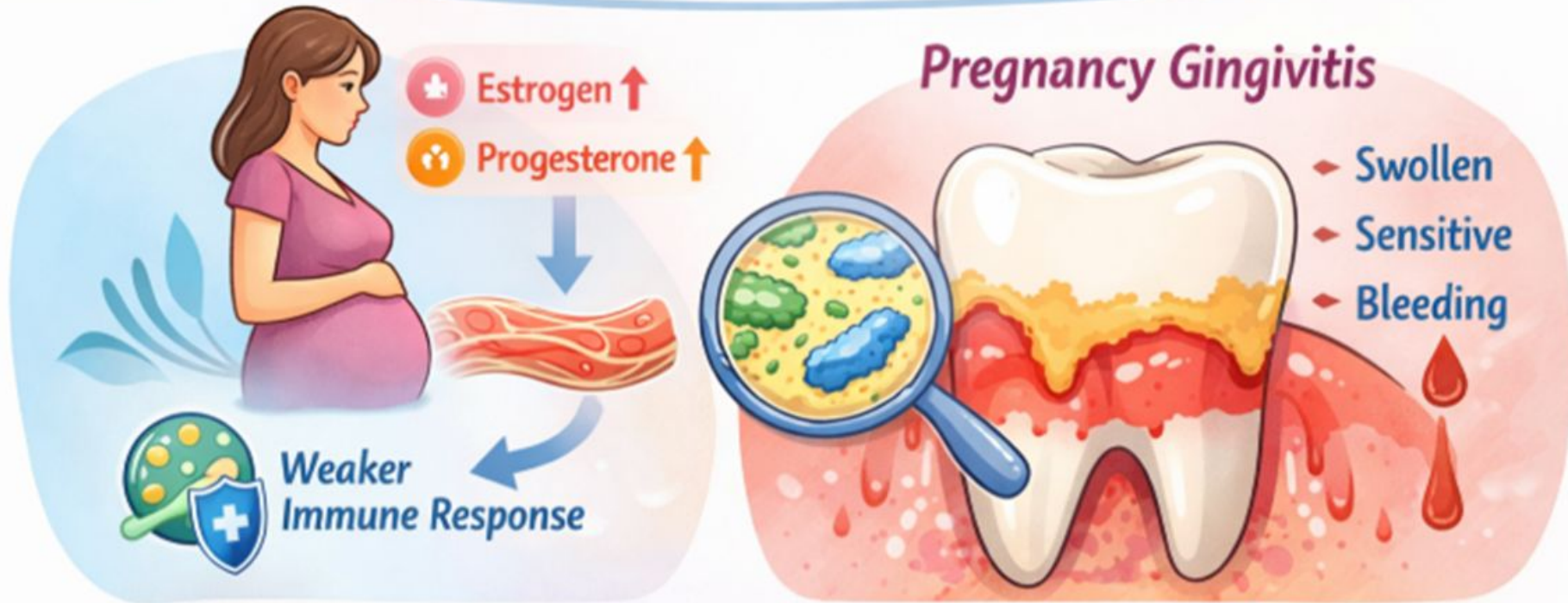
- Causes vasodilation
- Increases gingival edema
- Reduces collagen production
- Suppresses tissue repair
- Alters immune response

Result

Gingival tissues become more susceptible to inflammation and plaque irritation.

Why Gums Change During Pregnancy

Hormonal Changes & Pregnancy Gingivitis

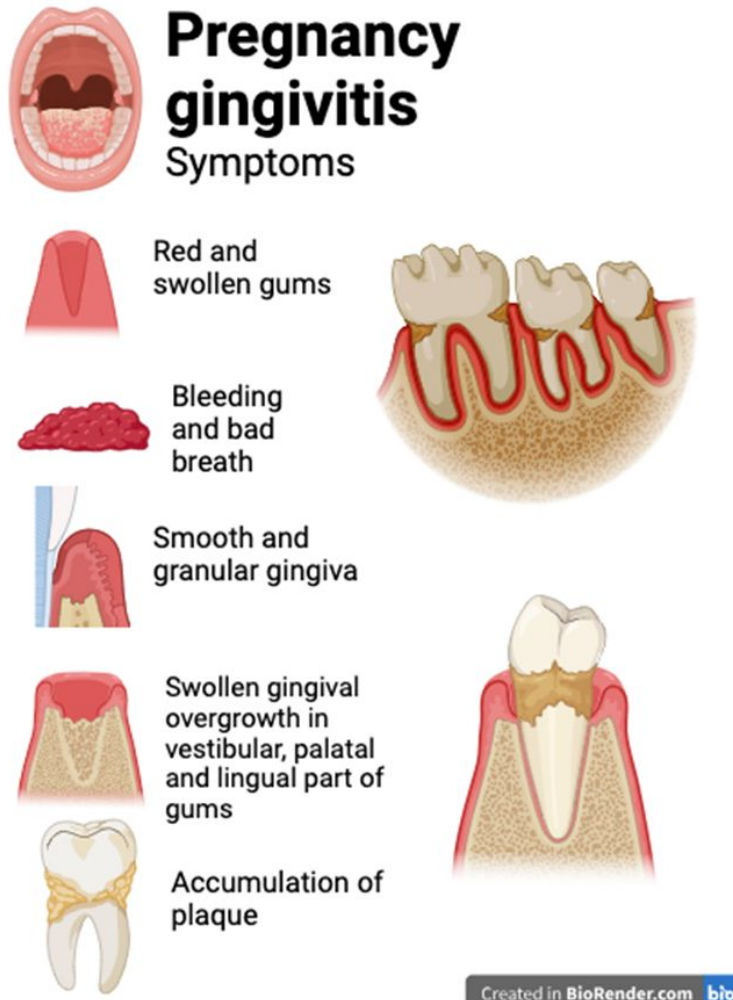


Plaque + Hormones = Increased Inflammation

Maintain Good
Oral Care



Hormonal Influence on Oral Tissues



Hormonal changes during pregnancy lead to:

- Increased blood flow to gingiva
- Tissue edema
- Increased inflammatory mediators
- Altered immune function
- Changes in oral microbial flora

Affected tissues include:

- Gingiva
- Oral mucosa
- Periodontal ligament
- Salivary glands

What is Pregnancy Gingivitis?



Definition

Pregnancy gingivitis is an inflammatory condition of gingiva occurring during pregnancy due to exaggerated gingival response to plaque under the influence of increased hormonal levels.

Characteristics

- Common during second and third trimester
- Reversible after childbirth
- Severity related to plaque accumulation

Epidemiology

- Occurs in approximately 60–75% of pregnant women.
- More common in women with poor oral hygiene.
- Severity increases with increasing pregnancy duration.
- Usually becomes prominent during the 2nd month and peaks in the 8th month.

Pathogenesis of Pregnancy Gingivitis

Mechanism

Step 1: Hormonal Increase

↑ Estrogen and progesterone

Step 2: Vascular Changes

- Increased blood vessel dilation
- Increased permeability

Step 3: Immune Alterations

- Reduced neutrophil efficiency
- Increased inflammatory response

Step 4: Plaque Interaction

Plaque bacteria trigger exaggerated inflammation.

Relationship Between Plaque and Hormonal Changes

Key Concept

Hormones alone do not cause gingivitis.

Plaque is the primary initiating factor.

However:

- Pregnancy hormones intensify gingival response to plaque.

Even small amounts of plaque may produce:

- Severe redness
- Swelling
- Bleeding

Thus, pregnancy acts as a modifying factor.

Dental Plaque



Definition

Dental plaque is a soft, sticky microbial biofilm attached to tooth surfaces.

Components

- Bacteria
- Salivary proteins
- Food debris

Effects

Plaque bacteria release:

- Toxins
- Enzymes
- Inflammatory substances

These initiate gingival inflammation.

Microorganisms Associated with Pregnancy Gingivitis

Important Bacteria

Prevotella intermedia

- Increases during pregnancy
- Uses progesterone and estrogen as growth factors

Porphyromonas gingivalis

- Associated with periodontal inflammation

Hormonal changes favor growth of these microorganisms.

Biochemical Basis of Gingival Inflammation

Inflammatory Mediators

Pregnancy gingivitis involves increased production of:

- Prostaglandins
- Cytokines
- Histamine

Important Cytokines

- IL-1
- TNF- α
- IL-6

These substances:

- Increase vascular permeability
- Cause tissue destruction
- Intensify inflammation

Changes in Collagen Metabolism

Role of Progesterone

Progesterone:

- Decreases collagen synthesis
- Inhibits fibroblast function
- Delays tissue repair

Result

- Weak connective tissue
- Increased gingival swelling
- Delayed healing

Clinical Features of Pregnancy Gingivitis



Signs and Symptoms

- Red gingiva
- Swollen gingiva
- Bleeding during brushing
- Tenderness
- Gingival enlargement
- Halitosis

Most commonly affects:

- Interdental papilla
- Marginal gingiva

Pregnancy Tumor (Pyogenic Granuloma)



Definition

Localized inflammatory gingival enlargement seen during pregnancy.

Features

- Red nodular mass
- Bleeds easily
- Soft consistency
- Rapid growth

Usually occurs:

- During second trimester
- On interdental papilla

Often regresses after delivery.

Histological Features

Microscopic findings:

- Dilated capillaries
- Edematous connective tissue
- Inflammatory cell infiltration
- Collagen breakdown
- Epithelial hyperplasia

These changes reflect chronic inflammatory reaction.

Pregnancy Gingivitis vs Periodontitis

Feature	Pregnancy Gingivitis	Periodontitis
Tissue affected	Gingiva only	Gingiva + supporting tissues
Bone loss	Absent	Present
Pocket formation	Minimal	Deep periodontal pockets
Reversibility	Usually reversible	Often irreversible
Main cause	Hormones + plaque	Chronic infection
Tooth mobility	Absent	May be present

Differentiation from Other Gingival Diseases

Pregnancy Gingivitis

- Occurs during pregnancy
- Hormone-related
- Exaggerated plaque response
- Usually reversible

Puberty Gingivitis

- Occurs during adolescence
- Due to pubertal hormonal changes

Drug-Induced Gingival Enlargement

Caused by:

- Phenytoin
- Cyclosporine
- Calcium channel blockers

Risk Factors

Factors increasing severity:

- Poor oral hygiene
- Existing gingivitis
- High plaque levels
- Nutritional deficiency
- Smoking
- Stress
- Diabetes mellitus

Importance of Oral Hygiene During Pregnancy

Good oral hygiene:

- Reduces plaque accumulation
- Prevents gingival inflammation
- Improves maternal oral health

Recommended Measures

- Brushing twice daily
- Flossing
- Antimicrobial mouthwash
- Regular dental visits

Dental Care During Pregnancy

Safe Dental Care Includes

- Scaling and polishing
- Oral hygiene instruction
- Preventive care
- Routine examination

Best Time for Dental Treatment

Second trimester is considered safest.

Prevention of Pregnancy Gingivitis

Preventive Strategies

Before Pregnancy

- Complete dental treatment
- Treat existing gingivitis

During Pregnancy

- Maintain plaque control
- Regular dental checkups
- Balanced nutrition
- Professional cleaning

Prevention is more effective than treatment.

Management

Treatment

Non-Surgical Therapy

- Plaque removal
- Scaling
- Root planing

Home Care

- Soft toothbrush
- Proper brushing technique
- Chlorhexidine mouthwash if advised

Surgical Management

Only for persistent pregnancy tumor.

Importance for Dental Professionals

Dentists should:

- Educate pregnant women
- Encourage preventive care
- Monitor gingival changes
- Provide reassurance
- Promote safe oral hygiene practices

Early intervention prevents complications.

Summary

- Hormones play a vital role in oral health.
- Pregnancy increases estrogen and progesterone levels.
- These hormones exaggerate gingival response to plaque.
- Plaque is essential for development of pregnancy gingivitis.
- Pregnancy gingivitis differs from periodontitis because there is no bone loss.
- Good oral hygiene and dental care are crucial during pregnancy.

Conclusion

Pregnancy gingivitis is a common inflammatory condition caused by interaction between hormonal changes and plaque accumulation. Elevated estrogen and progesterone alter vascularity, immunity, and connective tissue metabolism, making gingiva more sensitive to bacterial plaque. Proper oral hygiene, preventive dental care, and patient education are essential for maintaining oral health during pregnancy.

